NOTE THIS IS NOT SEVCON PIN COMPATIBLE. DO NOT CONNEC THIS TO SEVCON
IT WILL BLOW UP

J6
776231-1

BMS CAN1 GND -> CAN1_GND 1
BMS CAN1 LOW -> CAN1_LOW 2
BMS CAN1 HIGH -> CAN1_HIGH 3
BMS CAN1 +5V -> CAN1_+5V 4
5
REDUNDANT THROTTLE B INPUT -> THROTTLE2_IN_A 6
REDUNDANT THROTTLE A INPUT -> THROTTLE1_IN_A 7
THROTTLE GROUND -> GND_A 8
THROTTLE +5V -> +5V_A 9
CONNECT SWITCH BETWEEN +12V_A AND THIS PIN -> I_+12V_KEYSWITCH_A 10
BLANK TO ISOLATE TRACTION BATTERY VOLTAGE -> 11
TRACTION BATT+ (max 160v) -> I_+VIN_Z 12
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_A 13
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_B 14
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_A 15
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_B 16
AUX 12V OUT, TO DISPLAY, KEYSWITCH -> +12V_SWITCHED_A 17
USER DIGITAL INPUT 1 (see ctrl board for v/current limits) -> DIN_1_W 18
DIN GND -> GND_W 19
THROTTLE HOME POSITION SWITCH (REF'D TO THROTTLE +5V) -> I_+5V_IDLE_SWITCH_A 20

21 GND_A <- GND REFERENCE FOR MOTOR TMP, ENCODER
22
23 O_PRECHARGE_DC_LINK_+_Z <- CONNECT TO DC LINK+, ALLOWS INVERTER TO PRECHARGE DC LINK
24 CAN2_GND <- DASH, CHARGER, ABS CANBUS GND
25 CAN2_LOW <- DASH, CHARGER, ABS CANBUS LOW
26 CAN2_HIGH <- DASH, CHARGER, ABS CANBUS HIGH
27 CAN2_+5V <- DASH, CHARGER, ABS CANBUS +5V
28 MOTOR_TEMP_IN_A <- MOTOR THERMISTOR (TIE OTHER END OF THERMISTOR TO GND_A)
29 HALL_W_IN_A <- MOTOR HALL W (IF USED, IF SIN/COS LEAVE N/C)
30 HALL_V_ENCODER_COS_IN_A <- MOTOR HALL V OR ENCODER COSINE (DEPENDING ON SENSOR TYPE)
31 HALL_U_ENCODER_SIN_IN_A <- MOTOR HALL U OR ENCODER SIN (DEPENDING ON SENSOR TYPE)
32 +5V_A <- MOTOR ENCODER/TMP SENSOR +5V
33
34 O_+VIN_SWITCHED_Z <- OUTPUT, TIE TO SEVCON DC/DC ENABLE PIN
35 GND_Z <- TRACTION BATTERY MINUS TERMINAL
MH1 MH2 MH3 MH4

Connector to IO board
J7
Conn_02x08_Counter_Clockwise
HALL_U_ENCODER_SIN_IN_A 1 16 DOUT4_A
+5V_A 2 15 DOUT3_A
HALL_V_ENCODER_COS_IN_A 3 14 DOUT2_A
GND_A 4 13 DOUT1_A
HALL_W_IN_A 5 12 GND_A
GND_A 6 11 THROTTLE2_IN_A
MOTOR_TEMP_IN_A 7 10 GND_A
GND_A 8 9 THROTTLE1_IN_A

SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
Conn_02x08_Counter_Clockwise
GND_A 1 16 +12V_IN_A
MP_SWO 2 15 GND_A
MP_BOOTSEL 3 14 CP_BOOTSEL
MP_SWCLK 4 13 CP_SWCLK
MP_SWDIO 5 12 CP_SWDIO
MP_RESET 6 11 CP_RESET
GND_A 7 10 +3.3V_A
USB_DP 8 9 USB_DM

USB Debugging port
12Mbit/s - not powered
inverter must have external power
J1
UJ2-B-HR-G-V
MH2 GND_A
USB_DM 1 VBUS
2 D-
GND 4 GND_A
D+ 3 USB_DP
MH1
GND_A MH1

I_+5V_IDLE_SWITCH_A DIN_1_W
I_+12V_KEYSWITCH_A DIN_2_W
GND_A GND_W

Connector to IO board
J12
Conn_02x08_Counter_Clockwise
DIN_1_W 1 16 GND_W
GND_W 2 15 DIN_8_W
DIN_2_W 3 14 GND_W
GND_W 4 13 DIN_7_W
DIN_3_W 5 12 GND_W
GND_W 6 11 DIN_6_W
DIN_4_W 7 10 GND_W
GND_W 8 9 DIN_5_W

onboard isolated 40-160v -> 12v 20W power supply
GND_Z
INPUT POLARITY PROTECTION
F1
MF-RHT200_32-2
D3
S3N
D4
S3N
I_+VIN_Z
VIN_FUSED_Z
+12V_IN_A +12V_A
C189
100nF
PS3
EC7BW-110S12
REMOTE_ON/OFF
-VIN
+VIN
TRIM
-VOUT
+VOUT
GND_A
+12V_A

C47 C48 C49 C50 C51
C42 C43 C44 C45 C46
UCM1H101MCL1GS
GND_A +12V_A

+12V external switch
enables/disables aux 12v output.
Note, when system is running, this output must not be on
due to power constraints with onboard regulator
intended use case is to run aux systems while
traction power is not applied when trpwr is on,
offboard main power system should power loads instead.
DOUT4_A
U6
BTS462TATMA1
Vbb
IN
OUT
GND
+12V_A
+12V_SWITCHED_A
GND_A

Connector to IO board
J13
Conn_02x03_Counter_Clockwise
CAN2_JUMPER_A 1 6 CAN2_HIGH
CAN2_JUMPER_B 2 5 CAN2_LOW
CAN2_+5V 3 4 CAN2_GND
J10
Conn_02x03_Counter_Clockwise
CAN1_JUMPER_A 1 6 CAN1_HIGH
CAN1_JUMPER_B 2 5 CAN1_LOW
CAN1_+5V 3 4 CAN1_GND

IC10
DRDNB21D-7
DOUT1_A
GND_A
DOUT2_A
B2
K1/K2
B1
C2/A2
E1/E2
C1/A1
+5V_A
C188
100nF
GND_A
+5V_A

PRECHARGE CIRCUIT
K1
AWHSH105D00G
NOTE TO USER:
THEY MUST CONNECT EXTERNAL PRECHARGE RESISTOR
SUCH THAT <7A FLOWS
VIN_FUSED_Z
COIL_1
NO
NC
COIL_2
COM
+5V_A
O_PRECHARGE_DC_LINK_+_Z
K2
AWHSH105D00G
SEVCON DC/DC ENABLE CIRCUIT
VIN_FUSED_Z
O_+VIN_SWITCHED_Z

Prepared by Thomas Liao
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Sheet: /
File: MainBoard.kicad_sch
Title: Open Inverter Platform - IO Board
Size: USLetter | Date: 2026-07-07 | Rev: C
KiCad E.D.A. 10.0.3 | Id: 1/1

!!!TODO!!!